

Performance of RapidSmith Device Files

Part	XDLRC Report Size	RapidSmith File Size	Java Heap Usage*	Load Time In RapidSmith*
Virtex 4 LX15	0.7 GB	232 KB	39 MB	0.48 secs
Virtex 4 FX140	7.8 GB	1546 KB	140 MB	1.63 secs
Virtex 4 LX200	9.7 GB	1011 KB	116 MB	1.53 secs
Virtex 5 LX20T	0.9 GB	497 KB	39 MB	0.52 secs
Virtex 5 FX200T	9.2 GB	1227 KB	129 MB	1.62 secs
Virtex 5 LX330T	13 GB	1250 KB	153 MB	1.91 secs
Virtex 6 LX75T	3.0 GB	583 KB	54 MB	0.79 secs
Virtex 6 HX565T	18 GB	1659 KB	125 MB	1.81 secs
Virtex 6 LX760	23 GB	1756 KB	135 MB	2.48 secs

hard macro support with modules and module instances.

The packages in RapidSmith are device, design, util, placer, router, primitiveDefs and examples.

The RapidSmith device package. During installation, RapidSmith will parse the large XDLRC files to generate smaller but faster loading device file representations. These device files (see

table) are loaded automatically with a design. RapidSmith supports up to Virtex6 and comes with a graphical viewer to browse devices.

Design analysis tools. When a design is given in XDL format, RapidSmith provides an easy-to-use platform for creating design analysis tools. You can view primitive instances, pins and routing as well as search for

primitive instances in a loaded design. RapidSmith also has a design analyser that will tabulate detailed resource utilisation statistics and design properties.

Design creation tools. RapidSmith can also read and write XDL files as its API gives you a tool to create circuitry in the XDL format itself. There are API routines to create primitive instances, customise them and create logical connects between instance pins.

Physical design tools. RapidSmith also provides an excellent framework for creating physical design tools such as placement and routing tools.

What you can do with RapidSmith

RapidSmith allows rapid prototyping of custom tools and unique CAD algorithms. It can also create new kinds of placers, routers, mappers, packers, etc. Moreover, it provides several APIs to reach low-level parts of design in order to extract unique design information. ●

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